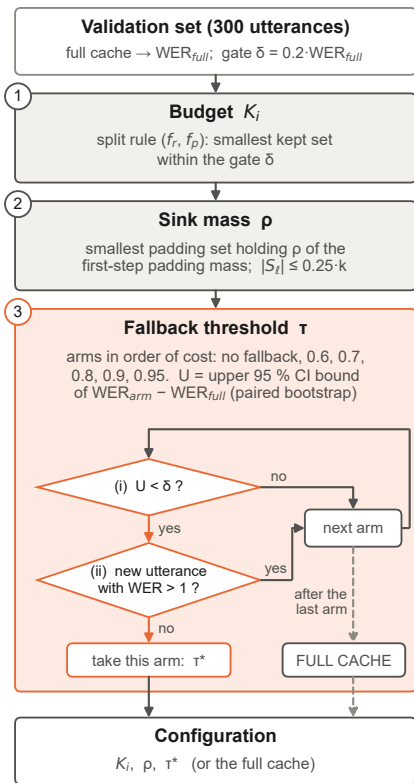


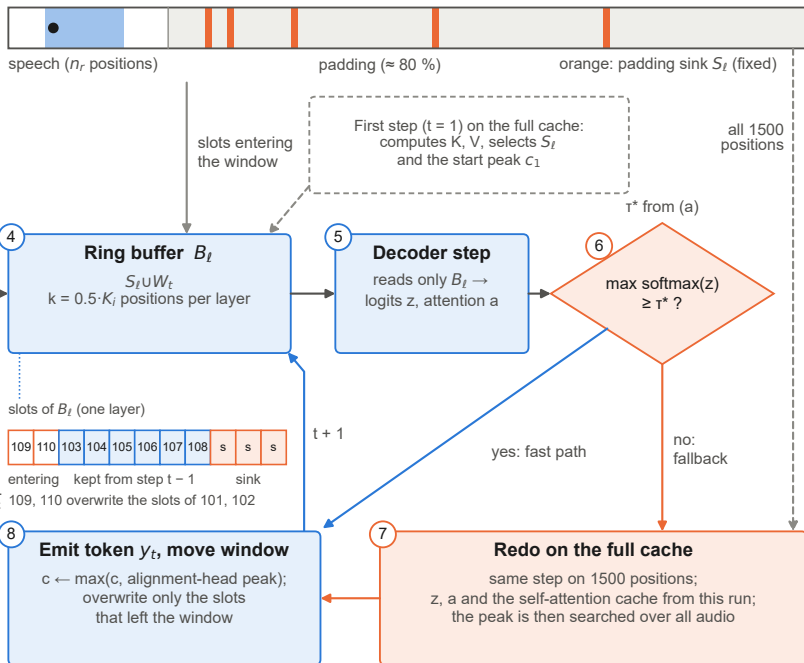
(a) Calibration, once per model



(b) Decoding step t

Full cross-attention cache K, V — 1500 positions per layer (stays in DRAM)

blue: window $W_t = [c_t - 0.1 \cdot w_t, c_t + 0.9 \cdot w_t]$, moves with decoding; • = alignment peak c_t



Legend

PadSink-Track fast path

padding sink / fallback

$\rightarrow$ every step / every arm

$\dashrightarrow$ conditional: first step, fallback, last arm

(c) What a step reads

fast path: k positions per layer (here $k = 196$, to scale)

fallback: 1500 positions per layer